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Q1. Harshad number

ANSWER

```
n=int(input())
temp=n
s=0
while temp>0:
    d=temp%10
    s=s+d
    temp=temp//10
if n%s==0:
    print("harshad no")
else:
    print("not harshad no")
for i in range(1,501):
    temp=i
    s=0
    while temp>0:
        d=temp%10
        s=s+d
        temp=temp//10
    if i%s==0:
        print(i)
```

Q2. GCD

CODE

```
a=int(input("enter first:\n"))
b=int(input("enter second:\n"))
while b!=0:
    temp=b
    b=a%b
    a=temp
print(a)
```

INPUT

```
24
18
```

OUTPUT

```
enter first:
enter second:
6
```

Q3. sum to a target value

ANSWER

```
def find_pairs(lst,target):
    seen=set()
    pairs=[]
    for num in lst:
        complement = target-num
        if complement in seen:
            pairs.append((complement,num))
        seen.add(num)
    return pairs
data=[int(x) for x in input("enter numbers:").split()]
t=int(input("target: "))
result=find_pairs(data,t)
print("pairs:",result)
```

Q4. Bitwise Operators

ANSWER

```

a=0b1101=13
b=0o25=21
c=0x1A=26
a=13=01101
b=21=00101
-----
a^b=00101=5
-----
5=00101
26=11010
-----
5|26=11111=31
result=31(int)
a=13,b=21,c=26,result=31
0b11111,0o37,0x1f
result>>2=7
result<<1=62
~result=-32

```

Q5. binary search

CODE

```

def rotated_binary_search(arr,target):
    low=0
    high=len(arr)-1
    while low<=high:
        mid=(low+high)//2
        if arr[mid]==target:
            return mid
        if arr[low]<=arr[mid]:
            if target >=arr[low] and target<arr[mid]:
                high=mid-1
            else:
                low=mid+1
        else:
            if target>arr[mid] and target<=arr[high]:
                low=mid+1
            else:
                high=mid+1
    return -1
data=[4,5,6,7,0,1,2]
t= int(input("target:\n"))
result=rotated_binary_search(data,t)
print("found at :", result)

```

OUTPUT

```

target:
found at : 4

```